Input data file for analysis and design using STAAD Pro is given here below:

```
STAAD SPACE HILINKW
START JOB INFORMATION
ENGINEER DATE 25-Nov-19
END JOB INFORMATION
INPUT WIDTH 79
PAGE LENGTH 15000
SET NL 1000000
UNIT METER KN
JOINT COORDINATES
1 0 0 3; 2 3.6 0 3; 3 0 0 0; 4 3.6 0 0; 5 -0.6 3 3.6; 6 0 3 3.6; 7 1.2 3 3.6;
8 2.4 3 3.6; 9 3.6 3 3.6; 10 4.2 3 3.6; 11 -0.6 3 3; 12 0 3 3; 13 1.2 3 3;
14 2.4 3 3; 15 3.6 3 3; 16 4.2 3 3; 17 -0.6 3 0; 18 0 3 0; 19 1.2 3 0;
20 2.4 3 0; 21 3.6 3 0; 22 4.2 3 0; 23 -0.6 3 -0.6; 24 0 3 -0.6; 25 1.2 3 -0.6;
26 2.4 3 -0.6; 27 3.6 3 -0.6; 28 4.2 3 -0.6;
MEMBER INCIDENCES
1 1 12; 2 2 15; 3 3 18; 4 4 21; 5 5 6; 6 6 7; 7 7 8; 8 8 9; 9 9 10; 10 11 12;
11 12 13; 12 13 14; 13 14 15; 14 15 16; 15 17 18; 16 18 19; 17 19 20; 18 20 21;
19 21 22; 20 23 24; 21 24 25; 22 25 26; 23 26 27; 24 27 28; 25 5 11; 26 11 17;
27 17 23; 28 6 12; 29 12 18; 30 18 24; 31 7 13; 32 13 19; 33 19 25; 34 8 14;
35 14 20; 36 20 26; 37 9 15; 38 15 21; 39 21 27; 40 10 16; 41 16 22; 42 22 28;
START GROUP DEFINITION
MEMBER
_COL 1 TO 4
_MAINT 28 to 30 37 TO 39
_MAINL 10 to 14 15 TO 19
_remB 5 to 9 20 to 27 31 to 36 40 to 42
END GROUP DEFINITION
SUPPORTS
1 TO 4 FIXED
DEFINE MATERIAL START
ISOTROPIC S355
E 2.05e+008
POISSON 0.3
DENSITY 77
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 35500 FU 510000 RY 1.5 RT 1.2
ISOTROPIC S275
E 2.05e+008
POISSON 0.3
DENSITY 77
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 27500 FU 430000 RY 1.5 RT 1.2
ISOTROPIC M40
E 3.33e+007
POISSON 0.2
DENSITY 25
ALPHA 1.2e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 40000
END DEFINE MATERIAL
MEMBER PROPERTY BRITISH
_remB TABLE ST TUB1001008.0
_col TABLE ST TUB25015012.0
28 to 30 37 TO 39 10 to 14 15 TO 19 TABLE ST TUB25025010.0
CONSTANTS
*BETA 45 MEMB 24 TO 26
*BETA 90 MEMB 1 TO 4
MATERIAL S355 ALL
LOAD 1 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
floor load
yrange 2.99 3.01 fload -1
LOAD 2 LOADTYPE Live  TITLE LL
floor load
yrange 2.99 3.01 fload -.5
LOAD 3 LOADTYPE Wind  TITLE WD
floor load
yrange 2.99 3.01 fload -.9
```

```
LOAD 4 LOADTYPE Wind  TITLE WU
floor load
yrange 2.99 3.01 fload .9
LOAD 5 LOADTYPE Accidental  TITLE VCNT
MEMBER LOAD
1 UNI GZ 100 1.25 1.75
LOAD 6 LOADTYPE Accidental  TITLE VCPNT
MEMBER LOAD
1 UNI GX 50 1.25 1.75
LOAD COMB 31 SLS1
1 1.0 2 1.0
LOAD COMB 32 SLS2
1 1.0 3 1.0
LOAD COMB 33 SLS3
1 1.0 4 1.0
LOAD COMB 51 ULS1
1 1.35 2 1.5
LOAD COMB 52 ULS2
1 1.35 2 1.5 3 0.75
LOAD COMB 53 ULS3
1 1.0 4 1.5
LOAD COMB 54 ULS4
1 1.0 5 1.0
LOAD COMB 55 ULS5
1 1.0 6 1.0
PERFORM ANALYSIS PRINT STATICS CHECK
LOAD LIST 31 TO 33
PRINT JOINT DISPLACEMENTS
PRINT SUPPORT REACTION ALL
LOAD LIST 51 TO 53
PRINT MEMBER FORCES ALL
*print force envelope ns 10 list 3 54 105 160 211 248
PARAMETER 1
CODE EN 1993-1-1:2005
* NA 0:orig, 1: BS, 4:NF, 7:SS, 9:MS
NA 7 ALL
* sgr 3: S235, 1: S275, 2: S355, 3: S420, 4: S460
SGR 2 ALL
*ly 4.4 member 64 to 67
*lz 4.4 member 64 to 67
*cmm 4 member 17 18
TRACK 2 ALL
fixed group
group mem 1 TO 4
group mem 28 to 30 37 TO 39
group mem 10 to 14 15 TO 19
group mem 5 to 9 20 to 27 31 to 36 40 to 42
CHECK CODE ALL
*SELECT OPTIMIZED
PERFORM ANALYSIS PRINT STATICS CHECK
CHECK CODE ALL
*PRINT ANALYSIS RESULTS
STEEL TAKE OFF
FINISH
```